UNIVERSITY OF KELANIYA – SRI LANKA
FACULTY OF SCIENCE

Bachelor of Science Honours in Management and Information Technology / Bachelor of Science Honours in Information Technology- Degree Examination, November / December 2025
Academic Year 2023/2024 – Semester II

Information Technology

INTE 22213/ INTE 22212(R) – Software Design Patterns and Frameworks

No. of Questions:4 **No of Pages:3** **Time: 2 1/2 hrs.**

*Answer **ALL** questions. When allocating time for questions, take the marks allocated into consideration. State any assumption that you make.*

*This is an **OPEN BOOK** exam.*

QUESTION 1 (25 Marks)

A supermarket introduces a self-checkout kiosk. Customers can scan items, apply optional add-ons (e.g., gift-wrap, eco-bag, express packing), and make payments. The kiosk operates in different states: *Idle*, *Scanning*, *Payment*, and *Completed*. The price of an item may have dynamic add-ons (gift-wrap, eco-bag, etc.) without creating subclasses for every combination. A single global tax calculator must be used across all checkout counters to maintain consistent tax computation.

1.1 Identify which design patterns are suitable for following cases by providing reasons. **[6 Marks]**

a) managing kiosk states.

b) applying optional add-ons to products and calculating final price dynamically.

c) ensuring only one tax calculator instance is used.

1.2 Define an interface **CheckoutItem** with methods: **getCost()** and **getDescription()**. **[5 Marks]**

1.3 Create a concrete class **GroceryItem** implementing **CheckoutItem**. **[5 Marks]**

1.4 Explain how you will incorporate the different kiosk states (Idle → Scanning → Payment → Completed) using a design pattern. Provide an example transition. **[5 Marks]**

1.5 Describe how the Singleton tax calculator can be integrated into the pricing flow. **[4 Marks]**

QUESTION 2 (25 Marks)

A Media Streaming Platform wants to implement a scalable and resilient system using a microservices architecture. The platform must support key features such as user profile management, secure content delivery, personalized recommendations, billing/subscription handling, and analytics. Each microservice should operate independently while ensuring seamless user experience.

Key functional requirements:

- **User Profile Management:** Register users, manage accounts, preferences, viewing history, and profiles.
- **Content Delivery:** Store, transcode, and stream video content using CDN support.
- **Recommendations:** Provide personalized movie/series recommendations using ML/AI models.
- **Billing & Subscriptions:** Manage subscription plans, payments, invoices, renewals.
- **Analytics:** Track streams, playback errors, user engagement, and generate insights.

Answer the following questions:

2.1 Draw a diagram to depict the above microservices scenario. **(5 Marks)**

2.2 How can an organization effectively implement and manage microservices architecture to achieve scalability, fault isolation, and continuous deployment for the above platform? **(5 Marks)**

2.3 Propose a suitable microservices architecture for the above media streaming platform. **(5 Marks)**

2.4 What methods can be used to maintain data consistency across the proposed microservices? **(5 Marks)**

2.5 How would you ensure security (user data, and service-to-service communication) in the proposed architecture? **(5 Marks)**

QUESTION 3 (25 Marks)

A library wants to automate its **book-borrowing workflow**. When a user requests a book, the system performs a sequence of checks:

Verify membership validity, Check book availability, Check pending fines
These checks must occur in order and be easy to extend.

Different categories of books such as *Fiction*, *Research*, and *Magazine* must be created depending on user request.

Some rare books are stored in a restricted archive. A user should not directly access these objects; instead, access should be controlled through verifies permissions.

Optional services like *Extended Borrow Period*, *SMS Notification*, and *Email Receipt* should be applied without creating subclasses for each combination.

3.1 Identify which design patterns match for the following scenarios by providing reasons.

a) sequential request-validation checks
b) creating different types of book objects
c) controlling access to rare books
d) adding optional services to the borrow transaction
[8 Marks]

3.2 Create a Book interface with two methods: getTitle() and getCategory(). **[2 Marks]**

3.3 Write a simple **BookFactory** outline that returns objects of **FictionBook**, **ResearchBook**, or **Magazine**. **[5 Marks]**

3.4 Explain how you would handle the membership check → availability check → fine check. **[5 Marks]**

3.5 Explain the method(s) controlling access to rare books and give one example. **[5 Marks]**

QUESTION 4 (25 Marks)

An e-commerce platform handles customer orders. Different payment methods (Credit Card, PayPal, Cash on Delivery) follow a **common workflow**:

1. Validate payment details
2. Process payment
3. Send receipt
   However, each payment method has variations in step 2 (processing).

The company wants to standardize the common workflow while allowing custom behaviour for each payment type.

When an order is placed, it goes through several automated checks (inventory check, fraud check, delivery-area validation). These checks must be maintainable and easily extendable.

A single global **OrderID generator** must ensure all orders receive a unique ID.

4.1 Which design patterns fit for following cases. Provide valid reasons.

a) structuring the payment workflow with common steps but different implementations
b) performing sequential automated validations
c) generating consistent global Order IDs
d) creating different payment objects
[8 Marks]

4.2 Define an abstract **PaymentProcessor** class using the appropriate design patten pattern with methods:

`validate()`, `processPayment()`, `sendReceipt()`, and `handlePayment()` (
[4 Marks]

3.3 Implement a concrete class **CreditCardPaymentProcessor** extending **PaymentProcessor**.
[3 Marks]

3.4 Briefly outline how a design pattern can be used for the validation flow (inventory → fraud → delivery area). **[5 Marks]**

3.5 Explain how you will design a OrderID generator and how it integrates with payment processing. **[5 Marks]**

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